

Explainable Stacked Ensemble Machine Learning for Multiclass Fetal Classification from Cardiotocography Data

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ABSTRACT

Cardiotocography (CTG) is commonly employed for intrapartum fetal monitoring; however, its interpretation frequently exhibits subjectivity and inter-observer variability. Machine learning techniques have demonstrated potential for automated fetal health classification; however, numerous models prioritize predictive accuracy at the expense of interpretability and sensitivity to minority classes. This study presents a stacked ensemble framework for multiclass fetal health classification (Normal, Suspect, Pathological) utilizing the UCI Cardiotocography dataset. The architecture uses Logistic Regression as a meta-learner and Random Forest, XGBoost, and Support Vector Machine as base learners. Using SMOTE in a stratified 10-fold cross-validation framework, class imbalance is fixed. The suggested ensemble gets 96.4% accuracy, 0.94 macro F1-score, and 0.92 recall for cases that are pathological. SHAP makes models easier to understand by finding clinically important indicators like abnormal short-term variability (ASTV). The suggested framework offers dependable and comprehensible AI-driven decision support for evaluating fetal health.

Keywords—Cardiotocography, Fetal Health Classification, Stacked Ensemble Learning, Explainable AI, SHAP, Clinical Decision Support.

1. INTRODUCTION

A. Background on Intrapartum Fetal Monitoring and Cardiotocography (CTG)

Monitoring the fetus during pregnancy and labor is crucial for averting negative perinatal outcomes. Cardiotocography (CTG) is a popular way to keep an eye on the fetal heart rate and uterine contractions during labor without having to touch the baby [2], [3]. However, visual interpretation of CTG signals is still subjective and can vary from person to person, making it hard to accurately find fetal distress.

B. Limitations of Existing Machine Learning Approaches

Automated CTG analysis has been extensively investigated using machine learning models.

Nonetheless, numerous studies primarily emphasize classification accuracy, providing minimal interpretability for clinical decision-making. Moreover, class imbalance in CTG datasets frequently diminishes the identification of pathological cases, which are clinically significant.

C. Research Objectives and Primary Contributions

This study proposes an interpretable stacked ensemble framework for multiclass fetal health classification (Normal, Suspect, Pathological) utilizing the UCI Cardiotocography dataset [1] to tackle these challenges. The method combines different types of machine learning models with training that takes into account imbalances and explainable AI techniques to make predictions more reliable and the clinical process more open. To the

best of our knowledge, this study is one of the first to integrate stacked ensemble learning, SMOTE-based imbalance management, and SHAP-based interpretability for multiclass fetal health classification utilizing cardiotocography data. The primary contributions of this study are as follows:

1. Stacked Ensemble Framework: Integration of Random Forest, XGBoost, and Support Vector Machine with Logistic Regression as a meta-learner.

2. Imbalance-Aware Learning: Application of SMOTE within stratified 10-fold cross-validation to improve detection of minority pathological cases.

3. Explainable AI Integration: Use of SHAP to provide interpretable feature-level explanations for model predictions.

4. Clinical Insight Generation: Identification of clinically relevant CTG indicators, particularly abnormal short-term variability (ASTV), as key predictors of fetal distress.

II. RELATED WORK

A. Clinical Foundations of Cardiotocography Data

Cardiotocography (CTG) is widely used for continuous fetal monitoring during labor, providing non-invasive assessment of fetal heart rate (FHR) and uterine contractions [3], [4]. Clinical interpretation typically follows standardized guidelines such as FIGO [2], yet remains subjective and susceptible to inter-observer variability. To improve diagnostic consistency, recent research has focused on extracting physiological indicators from FHR signals. Publicly available datasets, particularly the UCI Cardiotocography dataset [1], have enabled data-driven approaches for automated fetal health assessment.

B. Ensemble Machine Learning for Fetal Health Classification.

Machine learning methods have been increasingly applied to automate CTG interpretation [4], [5]. Early approaches, including Naïve Bayes classifiers, established baseline performance. More recent studies emphasize ensemble learning techniques, such as Random Forest and XGBoost, which demonstrate strong generalization performance and improved pattern recognition in CTG data [4], [5]. However, many existing studies prioritize overall accuracy while providing limited insight into model interpretability and minority-class detection.

C. Addressing Class Imbalance in Medical Datasets

Class imbalance is a common challenge in medical datasets, including CTG data where normal cases dominate pathological ones. This imbalance can bias classifiers toward the majority class and reduce sensitivity to high-risk fetal conditions [6]. Oversampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) are widely used to mitigate this issue [6]. Effective implementations apply oversampling within proper validation frameworks to prevent information leakage and ensure reliable performance evaluation.

D. The Role of Explainable AI (XAI) in Clinical Trust

The increasing complexity of machine learning models has raised concerns about their black-box nature in clinical applications [7]. Explainable AI techniques, particularly SHapley Additive exPlanations (SHAP), provide feature-level interpretation of model predictions [7]. Recent studies demonstrate that SHAP can reveal clinically meaningful CTG indicators and improve transparency in fetal health prediction models [7], [8]. Linking predictions with physiological markers such as short-term variability enhances clinician trust and interpretability.

E. Research Gap and Motivation

Despite these advances, existing studies often focus primarily on predictive accuracy while providing limited interpretability and inconsistent handling of class imbalance. In addition, few works combine ensemble learning, imbalance-aware training, and explainable AI within a unified framework for CTG analysis. To address this gap, this study introduces a stacked ensemble approach that integrates SMOTE-based imbalance handling with SHAP-driven explainability to achieve both reliable multiclass fetal health classification and clinically interpretable insights.

III. METHODOLOGY

This part talks about the proposed framework for using cardiotocography (CTG) data to automatically classify fetal health. The workflow includes preparing the dataset, preprocessing it, training it while being aware of imbalances, and making a stacked ensemble model that can be understood for predicting fetal health in multiple classes.

A. Dataset Description and Class Distribution

This study uses the publicly available UCI Cardiotocography dataset [1], which contains 2,126 recordings with 21 features derived from fetal heart rate and uterine contraction signals. The dataset categorizes fetal health into three classes: Normal, Suspect, and Pathological. As shown in Table I, the

dataset exhibits significant class imbalance. Normal cases constitute 77.8% of the data, while Suspect (13.9%) and Pathological (8.3%) represent minority classes.

TABLE I – CTG Dataset Class Distribution

Class	Instances	Percentage
Normal	1655	77.8%
Suspect	295	13.9%
Pathological	176	8.3%

Due to this skewed distribution, imbalance-aware learning strategies are necessary to ensure reliable detection of pathological fetal conditions. In the proposed framework, the dataset is processed through a structured pipeline that integrates preprocessing, SMOTE-based resampling, and stacked ensemble learning to improve predictive robustness while maintaining clinical interpretability.

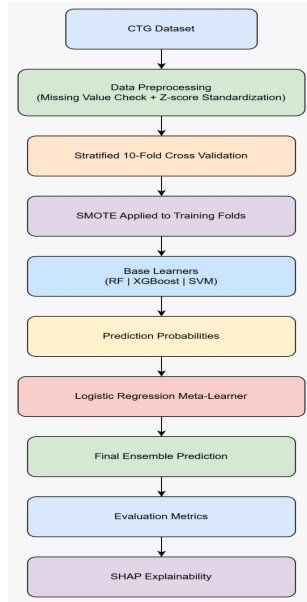


Fig. 1 illustrates the overall workflow of the proposed framework, including preprocessing, imbalance handling using SMOTE, stacked ensemble learning, and SHAP-based explainability.

B. Data Preprocessing and Feature Engineering

A preprocessing pipeline is applied to ensure reliable model training. The dataset is first examined for missing values and anomalies to maintain data integrity. Since algorithms such as Support Vector Machines (SVM) are sensitive to feature scale, z-score normalization is applied to standardize the feature distribution:

$$Z = (x - \mu) / \sigma.$$

where x represents the feature value, μ is the mean, and σ is the standard deviation. This transformation produces zero-mean and unit-variance features, preventing scale dominance during model training and improving learning stability.

C. Mitigating Class Imbalance with Stratified SMOTE

The CTG dataset shows a big difference between normal and abnormal fetal states. To solve this problem, the Synthetic Minority Over-sampling Technique (SMOTE) is used to make synthetic minority samples. To prevent information leakage, SMOTE is utilized exclusively on the training folds within a stratified 10-fold cross-validation framework, while the validation data remains unaltered. This design makes sure that evaluations are fair and that it is easier to find rare pathological cases.

D. Proposed Stacked Ensemble Architecture.

To enhance predictive robustness, a heterogeneous stacked ensemble architecture is developed by combining complementary learning algorithms with a meta-learning strategy.

1) Base Learners: Random Forest (RF), XGBoost, and Support Vector Machine (SVM) are used as diverse classifiers. RF improves robustness through bagging, XGBoost captures complex nonlinear patterns via boosting, and SVM with an RBF kernel models high-dimensional decision boundaries.

2) Meta-Learner: The outputs of the base learners are combined using Logistic Regression, which learns an optimal combination of model predictions:

$$y = f_{meta}(h_1(x), h_2(x), h_3(x))$$

where h_1 , h_2 , and h_3 represent predictions from RF, XGBoost, and SVM, respectively. This stacked design improves generalization by leveraging complementary model strengths while maintaining interpretability.

Algorithm 1: Stacked Ensemble Training Procedure

Input: CTG dataset $D(X, Y)$

Output: Trained stacked ensemble classifier

1. Perform preprocessing (missing value check and z-score normalization).
2. Apply stratified 10-fold cross-validation.
3. For each fold:
 - a. Use 9-fold for training and 1-fold for validation.
 - b. Apply SMOTE to the training folds only
4. Train base learners (RF, XGBoost, SVM)
5. Generate prediction probabilities from base learner
6. Concatenate predictions and train the Logistic Regression meta-learner.

7. Produce final ensemble predictions

E. Evaluation Metrics.

We use Accuracy, Precision, Recall, F1-score, and ROC-AUC to judge how well a model works. Because the CTG dataset has an uneven number of classes, the Pathological class's Recall and F1-score are given more weight. To measure how well the classification works, we use macro-averaged metrics and a confusion matrix. To explain the features, we use SHAP.

F. Hyperparameter Configuration

The hyperparameters used for the base learners and meta-learner are summarized in Table III.

TABLE III – HYPERPARAMETER SETTINGS

Model	Key Parameters
Random Forest	200 trees
XGBoost	150 estimators, LR=0.1, depth=6
SVM	RBF Kernel, C=1.0, gamma=scale
Logistic Regression	LBFGS solver, max_iter = 1000

IV. EXPERIMENTAL DESIGN AND EVALUATION FRAMEWORK

To evaluate the proposed stacked ensemble model, a structured experimental framework is implemented, including cross-validation, performance metrics, and comparison with baseline models.

A. Stratified 10-Fold Cross-Validation

All models are evaluated using stratified 10-fold cross-validation to preserve the class distribution of fetal health states (Normal, Suspect, Pathological). In each fold, 90% of the data is used for training and 10% for validation. To address class imbalance, SMOTE is applied only to the training folds, while validation data remains unchanged to prevent data leakage and ensure unbiased evaluation.

B. Performance Metrics.

Due to class imbalance in the UCI Cardiotocography dataset, evaluation emphasizes macro-averaged metrics and class-wise recall rather than accuracy alone. The following metrics are used:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \quad (2)$$

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \quad (3)$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^C \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (4)$$

Additionally, ROC-AUC evaluates discriminative capability across classification thresholds. Statistical significance between models is assessed using the Wilcoxon signed-rank test ($p < 0.05$).

C. Baseline Model Configuration.

The proposed framework is compared with three baseline classifiers: Random Forest, XGBoost, and Support Vector Machine. Random Forest uses 200 trees with the Gini criterion, XGBoost employs 100 estimators with a learning rate of 0.1 and depth 6, and SVM uses an RBF kernel with hyperparameters optimized via grid search.

D. Stacked Ensemble

The final model integrates the base learners using Logistic Regression as a meta-learner. To avoid information leakage, the meta-model is trained on out-of-fold prediction probabilities generated during cross-validation, enabling robust integration of heterogeneous model outputs.

E. Implementation Environment

All experiments were implemented in Python using the Scikit-learn and XGBoost libraries.

V. RESULTS AND ANALYSIS

This section presents the performance of the proposed stacked ensemble model for multiclass fetal health classification using CTG data, focusing on predictive accuracy, minority-class detection, and comparison with baseline models.

A. Overall Predictive Performance

The proposed stacked ensemble achieved 96.4% accuracy and a macro F1-score of 0.94, indicating strong and balanced classification performance across all fetal health categories.

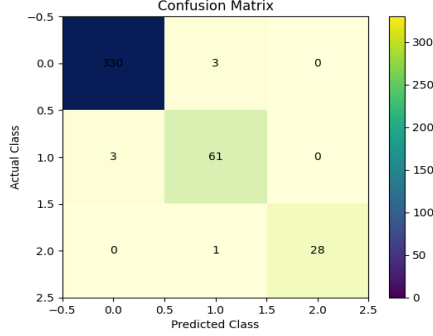
B. Minority-Class Sensitivity

Accurate detection of pathological fetal conditions is critical for clinical decision support. The proposed framework achieved a recall of 0.92 for the Pathological class, demonstrating effective identification of high-risk cases. This improvement

highlights the benefit of SMOTE-based imbalance handling within cross-validation.

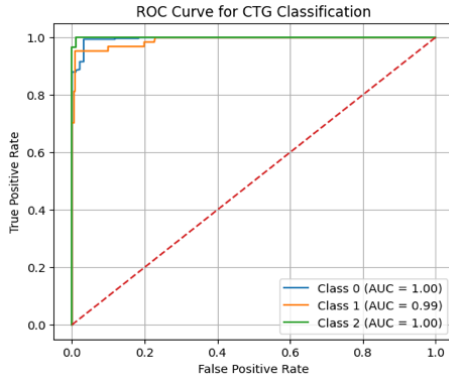
C. Confusion Matrix Analysis

Fig. 2 presents the confusion matrix of the proposed stacked ensemble across the three fetal health classes.



Most instances are correctly classified, with only a small number of pathological cases misclassified as Suspect, indicating reliable detection of clinically critical fetal distress.

D. ROC Analysis



The ROC curves in Fig. 3 illustrate the model's discriminative capability across classification thresholds. The proposed ensemble achieved an AUC of approximately 0.97, outperforming baseline models including SVM, Random Forest, and XGBoost.

E. Statistical Significance Testing

The Wilcoxon signed-rank test confirms that the performance improvements of the stacked ensemble are statistically significant ($p < 0.05$) compared to the baseline models.

F. Comparison with Baseline Models

The proposed ensemble outperformed the baseline classifiers, achieving the highest accuracy (96.40%)

and macro F1-score (0.94), while also improving recall for the minority Pathological class.

G. Feature Importance Analysis

Feature importance analysis indicates that key CTG indicators—baseline fetal heart rate, accelerations, uterine contractions, and decelerations—significantly influence the classification results.

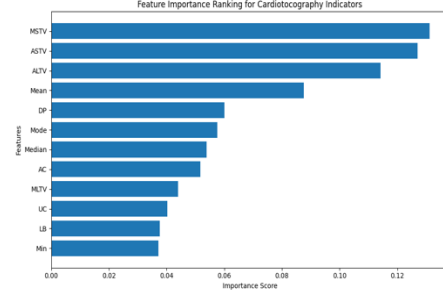


Fig. 4 illustrates the relative contribution of these features.

VI. EXPLAINABILITY AND CLINICAL VALIDATION

To improve model transparency, SHAP (SHapley Additive exPlanations) was used to interpret the stacked ensemble predictions.

A. Global Model Interpretability

The SHAP summary plot (Fig. 5) shows global importance. Abnormal short-term variability (ASTV) and prolonged decelerations emerge as the most influential predictors of pathological fetal health.

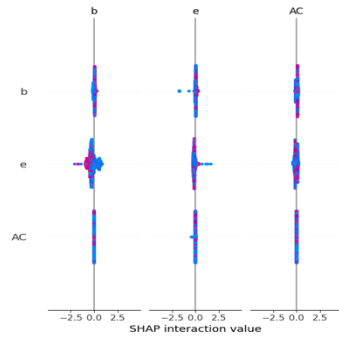


Fig. 5. SHAP summary plot showing the importance of key CTG features.

B. Local Interpretability

Local SHAP analysis explains individual predictions by showing how feature values influence model output. For high-risk cases, reduced short-term variability and abnormal heart rate patterns increase the probability of the pathological class.

C. Clinical Interpretation of Key Features

SHAP analysis highlights clinically relevant CTG indicators, including abnormal short-term variability, prolonged decelerations, and accelerations, supporting the clinical validity of the proposed framework.

VII. DISCUSSION

The proposed framework demonstrates the effectiveness of interpretable machine learning for fetal monitoring. By combining stacked ensemble learning with SMOTE-based imbalance handling, the model achieved strong detection of pathological cases (recall = 0.92). SHAP-based explanations link predictions to clinically relevant indicators such as abnormal short-term variability (ASTV) and prolonged decelerations, improving model transparency. However, the study is limited by the relatively small UCI CTG dataset and reliance on synthetic oversampling. Additionally, using pre-extracted features instead of raw CTG time-series signals may restrict the capture of temporal patterns.

VIII. CONCLUSION AND FUTURE DIRECTIONS

This study introduces a comprehensible stacked ensemble framework for multiclass fetal health classification utilizing CTG data. The proposed model attained an accuracy of 96.4%, a macro F1-score of 0.94, and a recall of 0.92 for pathological cases, indicating dependable identification of high-risk fetal conditions. SHAP-based analysis verified that the model utilizes clinically significant indicators, enhancing transparency for clinical decision support. Future endeavors will concentrate on validation through extensive multi-center datasets and the investigation of deep learning models that directly analyze CTG time-series signals.

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